

Product Rule Explanation

The Product Rule: General Formula

If you have a product of two functions $f(x) \cdot g(x)$, the derivative is:

$$\boxed{\frac{d}{dx}[f(x) \cdot g(x)] = f'(x) \cdot g(x) + f(x) \cdot g'(x)}$$

In words: First function times derivative of second, PLUS second function times derivative of first.

Memory trick: “First times derivative of second, plus second times derivative of first”

Simple Example

Find $\frac{d}{dx}[x^2 \cdot \sin(x)]$

- First function: $f(x) = x^2$, so $f'(x) = 2x$
- Second function: $g(x) = \sin(x)$, so $g'(x) = \cos(x)$

Apply product rule:

$$\begin{aligned}\frac{d}{dx}[x^2 \cdot \sin(x)] &= 2x \cdot \sin(x) + x^2 \cdot \cos(x) \\ &= 2x \sin(x) + x^2 \cos(x)\end{aligned}$$

Problem 4 Application

In problem 4, we needed to differentiate the entire expected utility function:

$$EU(p) = (1 - p)u(w_0) + p \cdot u\left(w_0 - \frac{R}{p}\right)$$

The first term is straightforward:

$$\frac{d}{dp}[(1 - p)u(w_0)] = -1 \cdot u(w_0) = -u(w_0)$$

The second term requires the product rule AND chain rule.

Applying Product Rule to the Second Term

For $\frac{d}{dp} \left[p \cdot u \left(w_0 - \frac{R}{p} \right) \right]$:

- First function: $f(p) = p$, so $f'(p) = 1$
- Second function: $g(p) = u \left(w_0 - \frac{R}{p} \right)$, so $g'(p) = ?$ (need chain rule)

Product rule gives:

$$\frac{d}{dp} \left[p \cdot u \left(w_0 - \frac{R}{p} \right) \right] = \underbrace{1 \cdot u \left(w_0 - \frac{R}{p} \right)}_{\text{first' times second}} + \underbrace{p \cdot \frac{d}{dp} \left[u \left(w_0 - \frac{R}{p} \right) \right]}_{\text{first times second'}}$$

Now Apply Chain Rule for $g'(p)$

We need $\frac{d}{dp} \left[u \left(w_0 - \frac{R}{p} \right) \right]$:

- Outer function: $u(w)$, so derivative is $u'(w)$
- Inner function: $w_0 - \frac{R}{p}$, so derivative is $\frac{R}{p^2}$

Chain rule:

$$\frac{d}{dp} \left[u \left(w_0 - \frac{R}{p} \right) \right] = u' \left(w_0 - \frac{R}{p} \right) \cdot \frac{R}{p^2}$$

Putting It All Together

$$\begin{aligned} \frac{d}{dp} \left[p \cdot u \left(w_0 - \frac{R}{p} \right) \right] &= u \left(w_0 - \frac{R}{p} \right) + p \cdot u' \left(w_0 - \frac{R}{p} \right) \cdot \frac{R}{p^2} \\ &= u \left(w_0 - \frac{R}{p} \right) + u' \left(w_0 - \frac{R}{p} \right) \cdot \frac{R}{p} \end{aligned}$$

Complete Derivative of EU

Therefore:

$$\begin{aligned} \frac{dEU}{dp} &= \frac{d}{dp} \left[(1-p)u(w_0) + p \cdot u \left(w_0 - \frac{R}{p} \right) \right] \\ &= -u(w_0) + u \left(w_0 - \frac{R}{p} \right) + u' \left(w_0 - \frac{R}{p} \right) \cdot \frac{R}{p} \end{aligned}$$

More Examples

Example 1: $\frac{d}{dx}[x^3 e^x]$

$$\begin{aligned}\frac{d}{dx}[x^3 e^x] &= 3x^2 \cdot e^x + x^3 \cdot e^x \\ &= e^x(3x^2 + x^3)\end{aligned}$$

Example 2: $\frac{d}{dx}[(2x + 1)(x^2 - 3)]$

$$\begin{aligned}\frac{d}{dx}[(2x + 1)(x^2 - 3)] &= 2 \cdot (x^2 - 3) + (2x + 1) \cdot 2x \\ &= 2x^2 - 6 + 4x^2 + 2x \\ &= 6x^2 + 2x - 6\end{aligned}$$

Example 3: $\frac{d}{dx}\left[x \cdot \frac{1}{x+1}\right]$

$$\begin{aligned}\frac{d}{dx}\left[x \cdot \frac{1}{x+1}\right] &= 1 \cdot \frac{1}{x+1} + x \cdot \left(-\frac{1}{(x+1)^2}\right) \\ &= \frac{1}{x+1} - \frac{x}{(x+1)^2} \\ &= \frac{x+1}{(x+1)^2} - \frac{x}{(x+1)^2} \\ &= \frac{1}{(x+1)^2}\end{aligned}$$

When to Use Product Rule

Use the product rule whenever you have:

- Two separate functions multiplied together
- NOT a constant times a function (just use constant multiple rule)
- NOT a composition like $f(g(x))$ (that's chain rule)

Example of when NOT to use it:

- $\frac{d}{dx}[5x^2]$: This is just $5 \cdot 2x = 10x$ (constant multiple)
- $\frac{d}{dx}[\sin(x^2)]$: This needs chain rule, not product rule